

Human Development Index (HDI) Predictor

Define Problem Statement

Project: Human Development Index (HDI) Predictor

Phase: Brainstorming & Ideation

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1. Background

The Human Development Index (HDI) is the standard measure used by the UN and researchers to compare countries on health, education, and income together, rather than economic output alone. Computing it, however, still requires manually gathering three separate indicators per country and applying the UNDP composite formula, which is slow when someone wants a quick estimate for a hypothetical or partial set of indicators.

2. Problem Statement

"How can we build a machine learning system that predicts a country's Human Development Index score directly from its Life Expectancy, Mean Years of Schooling, GNI per Capita, and Internet Users, and classifies it into a development tier, without requiring a manual recomputation of the UNDP formula?"

3. Root Cause Analysis

- HDI's official formula (geometric mean of three sub-indices) is not intuitive to recompute by hand
- Indicator data for a country is often scattered across multiple reports/sources
- No lightweight, interactive tool exists for a quick what-if HDI estimate
- Comparing many countries manually is repetitive and error-prone

4. Impact of the Problem

- Researchers and students spend extra time manually deriving HDI-like estimates
- Policymakers lack a fast way to explore "what if this indicator improved" scenarios
- Without a simple tool, HDI insights stay locked in annual static reports

5. Proposed Solution

Build an end-to-end ML pipeline that loads a country-indicator dataset, performs EDA and preprocessing, trains a Linear Regression model to predict HDI, and deploys it through a Flask web application that returns an instant HDI score and development category for any given set of indicator values.

6. Success Criteria

- Model achieves R^2 above 0.90 on held-out test data
- Predictions are sensible across the full development spectrum (Low through Very High)
- Web application accepts indicator input and returns an instant HDI score and category
- Null/missing indicator values are handled without breaking the pipeline